Final year (& Year 3 MEng) Projects:

Our Poster and Bench inspection day

One of the later stages of your Final Year (or Year 3 MEng) Project is a Poster presentation and Bench inspection of your ongoing project work. Since projects may not be quite finished yet, it is appreciated that the Poster and Bench inspection will show what you have done so far and any likely work in progress. The inspection is to take place at the bench in the Final Year Laboratory on the day that has been allocated to you. Assessment will take place throughout the day and will form an important part (20%) of the final mark for your project.

You will need to be present at your designated bench from **9.00 am** onwards throughout the day Early in the day the safety team will inspect hardware projects (please have a copy of your risk assessment at hand). Unsafe parts will be removed from the labs. Throughout the day, be available to present and explain in detail your project work to your Supervisor and Assessor who will **both** mark you. Additionally, you may also need to present your work to the External Examiners and the visiting Industrial Advisory Board. You should be **very well prepared** to demonstrate your project output (e.g., any designs, demonstrations, programs, circuits or other main results) and summarise your work. Moreover, you should be able to **answer any questions** related to your project work, aims and objectives, deliverables, or the general research area, and the industrial relevance and general applicability of your work. A record of your work undertaken so far (e.g. Preliminary Report or any other reports) and your log book should also be available for inspection.

The laboratory technicians will produce a laboratory layout showing everyone's bench locations. Please, confirm as soon as possible with the FY Lab Technicians that your project position is indicated on the Laboratory Layout otherwise another student may be allocated to have their bench inspection at that location and you will have to move. **The Lab Technicians must know where all students will be during the allocated day of bench inspection**.

The Poster should be standing near your Bench location and must be mounted on an A1-size posterboard (available from the lab technicians). Guidelines for the preparation of your Poster are given on the next page; further advice on presenting specific project details and aspects may be obtained from your Supervisor. Examples of posters from previous years are available in the Final Year Lab.

For those of you intending to print a poster through CSD using PowerPoint, the following guidelines about printing posters are available at: http://www.liv.ac.uk/csd/printing. Make sure to allow plenty of time for CSD to print your poster. **Do not leave it to the last minute!** The laboratory is currently open for you to make preparations at your bench, and it is **strongly recommended** that you begin your preparations for this week and finalise them before the bench inspection day. If your allocated day is Thursday then you should clear your bench at the end of the day so that students allocated to the following day can set up their posters without delay.

Around a week before Bench Inspection, we also ask for two online submissions. You will need to submit a **project abstract** the week before the bench inspection. This is a ~200 word, text-only summary of your project and will mostly be used to inform external visitors (external examiners and Industrial advisory board). For record-keeping, we also ask you to **submit a digital copy of your poster** online. You will only be marked using the paper copy on the day and submission will be open until after the bench inspection.

Poster Preparation Guidelines

1. **Size:** A1 mounted on poster board.

2. **Title:** Clearly indicate your Project Title, Student name (or names for Group Projects), Supervisor's name, and Assessor's name.

3. **Introduction/overview**: Clearly state what the project is about, why is it relevant to industry and technology and how important it is in the field. State clearly aims and motivations. Real-world and industrial relevance and applicability must be mentioned. The student must be able to elaborate on all these elements in detail at the end of the presentation during the questions stage.

4. **Initial and intermediate designs:** Present elements of the underlying theory, circuits or mathematical models produced, experiments undertaken, investigations carried out, program designs, etc. Give an overview of how you have attempted your project work (methodology). You may use text (but do not overdo it with too much text with small fonts), graphs, diagrams, flowcharts, equations, photos, etc.

5. **Ongoing designs:** Show how you have achieved the project goals so far, and state any goals to be achieved in the final stages of your work.

6. **Project output so far:** Give an interpretation of results, a discussion of performance/accuracy, whether your circuit/model/experiment produced the expected output, and if not why. It is also appropriate to discuss any difficulties you have experienced in this section. **How does this compare to your original specifications**?

7. **Conclusions:** Summary of what you have done, learned, produced or achieved so far. You should include here your proposals for further work or subsequent steps.

8. **Layout:** Possible layouts of your poster could be: a banner title (and names) with no more than 6 A4 pages, or a single large A1-sized PowerPoint slide to fit all of the above elements.

Lab book

Your lab book will be marked at the Bench Inspection so it is of crucial importance that you bring it along.
The lab book is a key element of your project. In industrial environments, lab books are used to log all preliminary data, calculations, initial results, etc. it should be kept as a chronological record as your project progresses and evidence the work you have done. Good record keeping should allow for reproducing experiments, tracking errors or helping to explain why artefacts occurred. Since lab books are record as you work, they don't need be excessively neat or tidied up afterwards or should never be written retrospectively.
Lab books can be kept on paper or digital

In a good lab book
All preliminary work is presented in the lab book.
All work is clearly dated.
Results are recorded in detail, allowing to reconstruct experiments.
Result graphs, source files etc. are clearly referenced.
Detailed observations are noted in the lab book.

Key points and some advice for the event

1) The students must be ready by **9:00**. All students must remain in the lab until they have been **marked by both the Supervisor and Assessor**. Due to the large number of projects, many students could be marked later on in the day until **17:00**. You may make arrangements with your project Supervisor and Assessor to set a specific time with him/her on the day to assess your project work.

2) Even if your Supervisor and Assessor visit you early, remain in the lab at your bench location, as our **External Examiners (EE), the Industrial Advisory Board (IAB) and other external visitors** will be visiting our department, and they will also be assessing many of the projects they wish to examine (but not all).

3) Please, **be patient** until supervisors, assessors, and perhaps external examiners and industrial advisors visit your bench. **Do not leave your post** unless it is absolutely necessary, and in this case, ask one of your fellow students to watch over your bench in case you are visited during that time.

4) Keep **noise** and other disruptions to a minimum, especially when one examination is taking place near you. Do **not** make telephone calls.

5) You must **prepare extremely well** and be ready with your poster and software/hardware demonstration and not leave anything at all to the last minute.

6) The quality of your presentation will form part of your mark. Practice how you will present your achievements. A video can be helpful to demonstrate how your project works. Consider time laps videos for experiments that take too long to demonstrate or animations to show simulation results. A video can also be a great backup in case your device does not work properly on the day.

7) Contact the lab technicians to finalise your bench location as soon as possible. You must have a **bench number** allocated well in advance.

8) There will be a **lunch break** from 1 p.m. to 2 p.m. After that please return to your bench **immediately**.

9) Even if you have produced high-quality deliverables in your project, it is very important that you can explain (to assessors, examiners and industrialists) the **industrial relevance** of your project and be able to justify its importance in today's technological **world** and its potential impact on our **society**. Therefore:
    a. **Study** the background and rationale of the work you do within its broader field and context. Many scientific papers and books have such information in their introductions.
    b. Also, **ask** your supervisors to help you with this, and explain to you their views on the matter. Actually, you should have already done this, as part of your work during the year.
    c. **Think** and analyse the wider context of your work, and how it could be applied.
    d. How do your achievements relate to the original **project specifications?** What has changed during the project? What was easier/more difficult than anticipated? How have you overcome problems?

10) At the end of your presentation there will be questions posed to you. **Listen carefully** till the end of the question and do not hastily reply.

11) All projects (and especially the ones based on simulation, software and theory) must be able to demonstrate how much **time** and **effort** was invested in the project. Make sure you show this in the presentation, the poster and perhaps in the questioning session.

12) For those projects that have not created a piece of hardware, that is the **design/simulation/software** projects, make sure you demonstrate they have been experimentally validated and the employed mathematics and algorithms have produced **experimentally validated results**. Even if the visitors have no time to see them, offer to demonstrate if they wish, or at least state the experimental status of your work.

13) Consider a video demonstration of your project. Videos can be a great way of demonstrating your result in particular if your demonstration might take too long, has aspects that cannot be shown in the lab or if something doesn't work quite right last minute.

14) The different assessors, examiners and industrialists will be asking you **diverse questions**. Make sure you are prepared for questions that **target different aspects** of your work. Some possible examples of non-technical questions may include:
    a. "Why is your project useful to society and how can it help our world to improve?"
    b. "What is the state-of-the-art of previous attempts to solve the problem you are tackling?"
    c. "Why did you decide to undertake this project?"
    d. "What was the most challenging part of your work?"
    e. "Your project seems to be a simulation or math based (i.e., non-hardware) one; what experimental and validation procedures did you use and why?"
    f. "How does your project fit in the general area of engineering and technology?"
    g. "This is a good project! I am sure you have been curious enough to find you how it could be developed further. Please, explain to me."
    h. "Could your project be used now by some industry to create a product or be part of a product? What industry? Mention some national or international industries that could be interested in your work."
    i. "How did you test your hardware (or software) system and its different modules to verify that it works, and how did you compare it with existing systems?"
    j. "What do the results show? Was your project successful? Does it compare well with competitors' systems? Why did you plan it the way you did?"
    k. "What is the originality of your project? Has anybody attempted something similar before? Why or why not?"

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